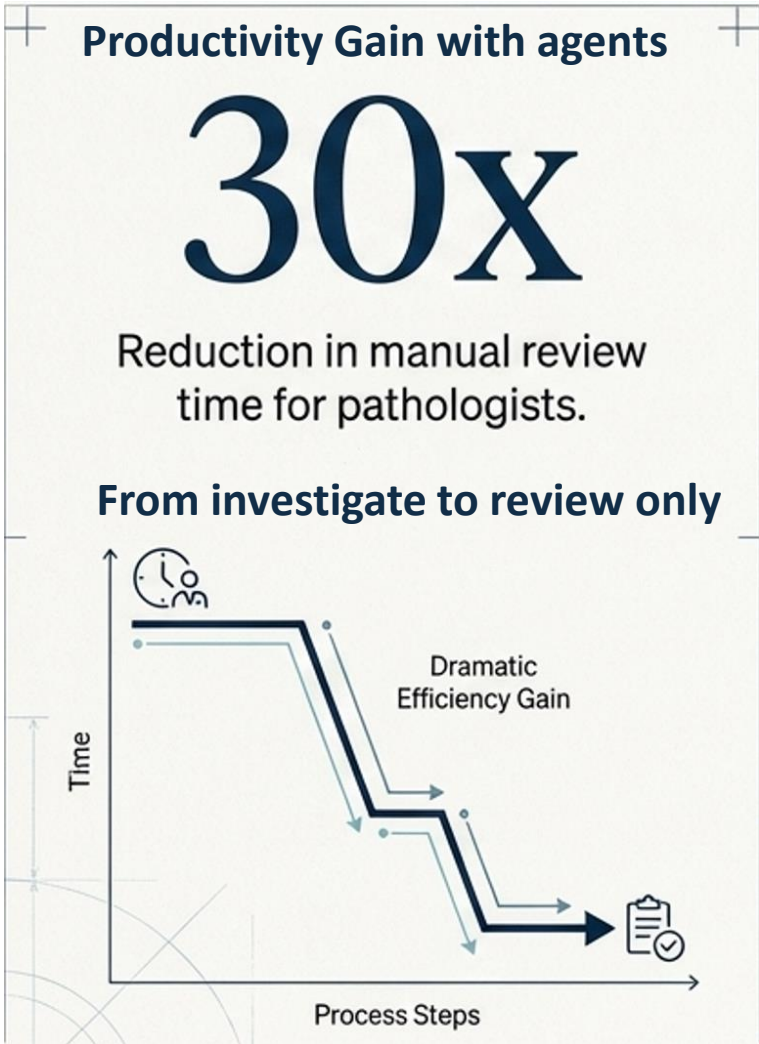
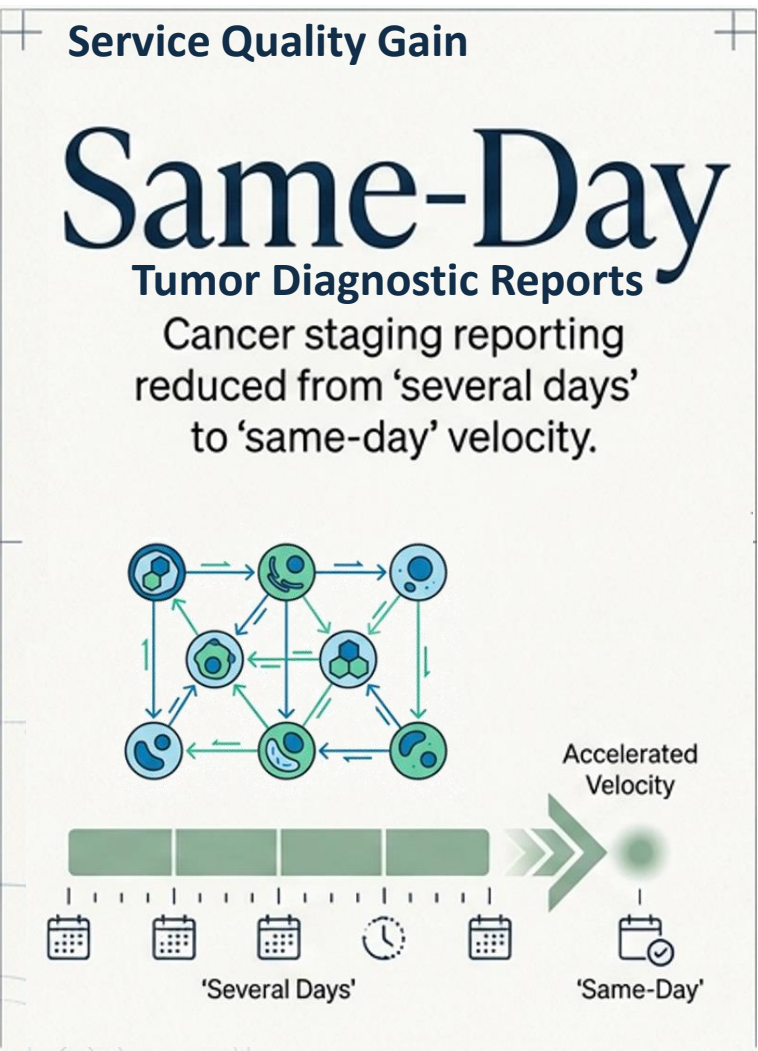
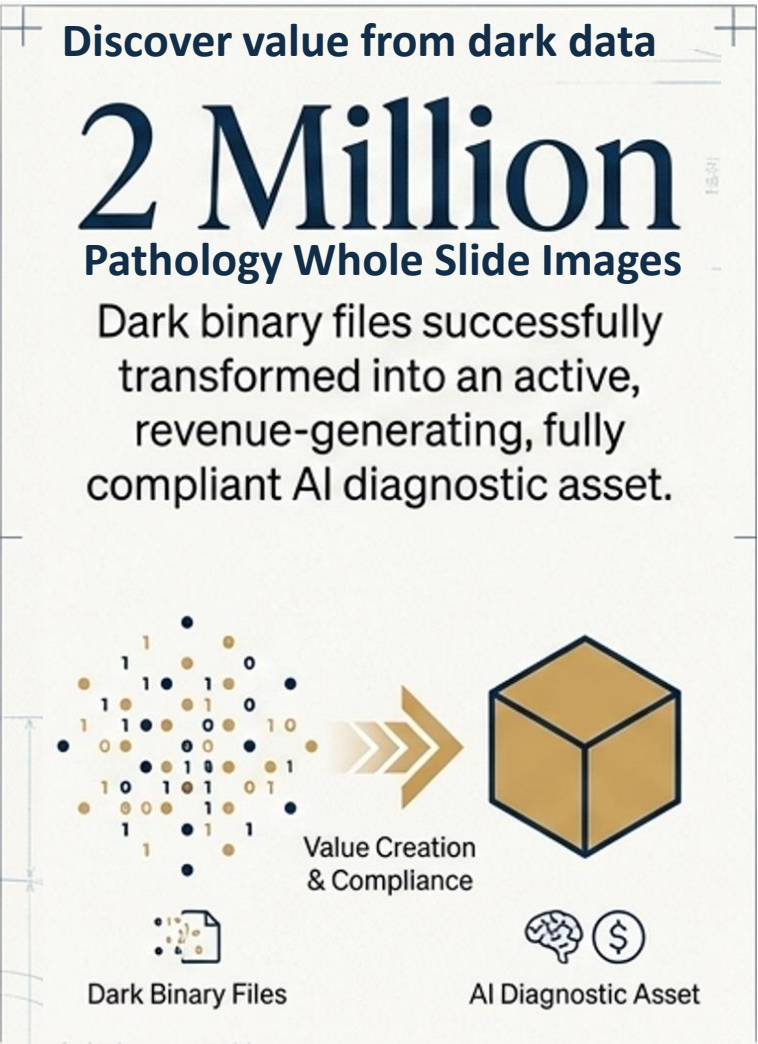


Database Modernization for Digital Pathology with Clinical Foundation Models

This is a story about

Turning dark data from a regulatory liability into an operational asset

Resulting in ...



From Dark Data to Diagnostic Insight:

The RuiPath Transformation at Shanghai Ruijin Hospital

The Challenge: A Legacy of “Dark Data”



Manual Microscope Review

Pathologists were forced to manually review every slide, a process prone to inconsistency and significant diagnostic delays.



The Labeling Bottleneck

The hospital had zero ground-truth labels for tumor stages, making traditional supervised AI training impossible without massive manual effort.

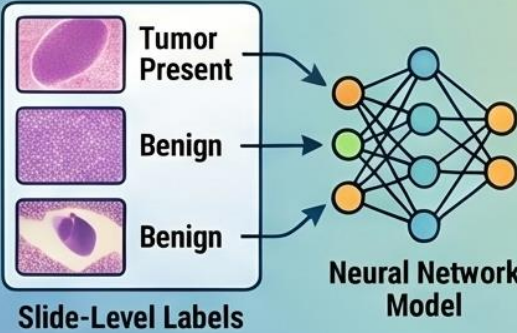
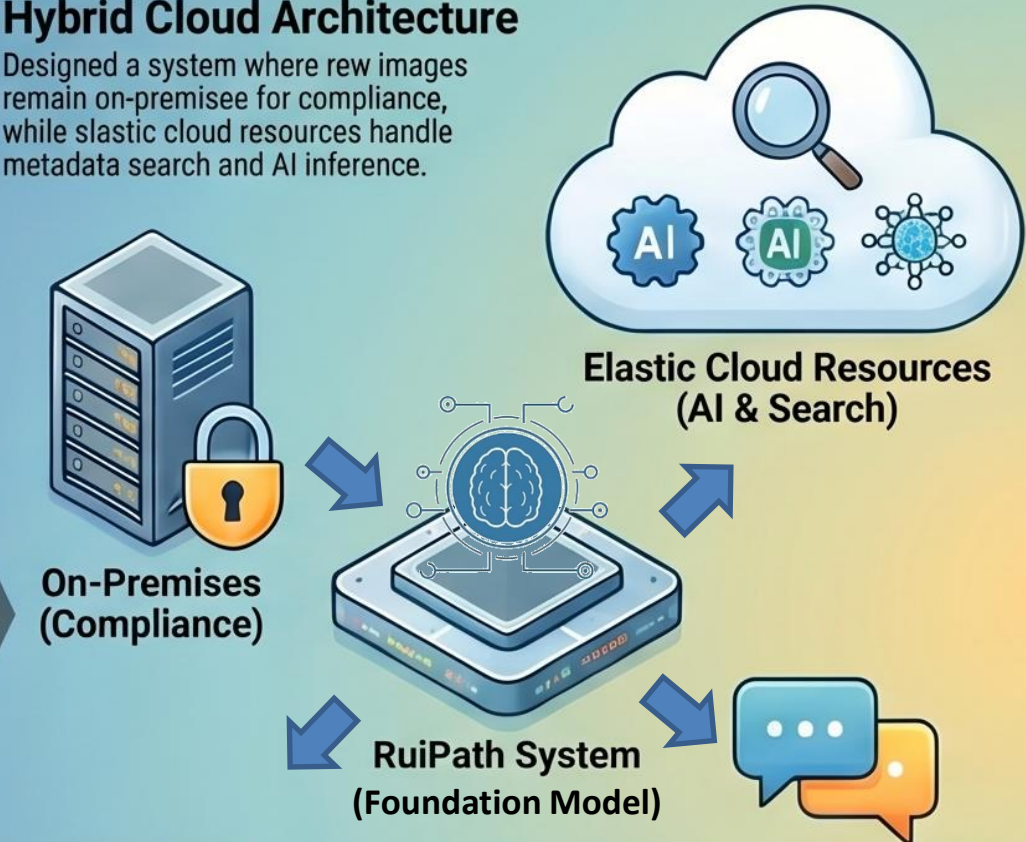
2 Million
“Dark” Images

Over a decade of high-resolution whole-slide images were stored as unstructured binary files, inaccessible for AI or machine learning workloads.

The Transformation: Building RuiPath

Hybrid Cloud Architecture

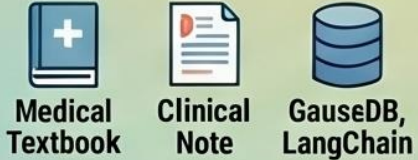
Designed a system where raw images remain on-premise for compliance, while elastic cloud resources handle metadata search and AI inference.



Multi-Instance Learning (MIL)

Implemented “weak supervision” to train models using only slide-level labels from pathologists rather than pixel-level annotations.

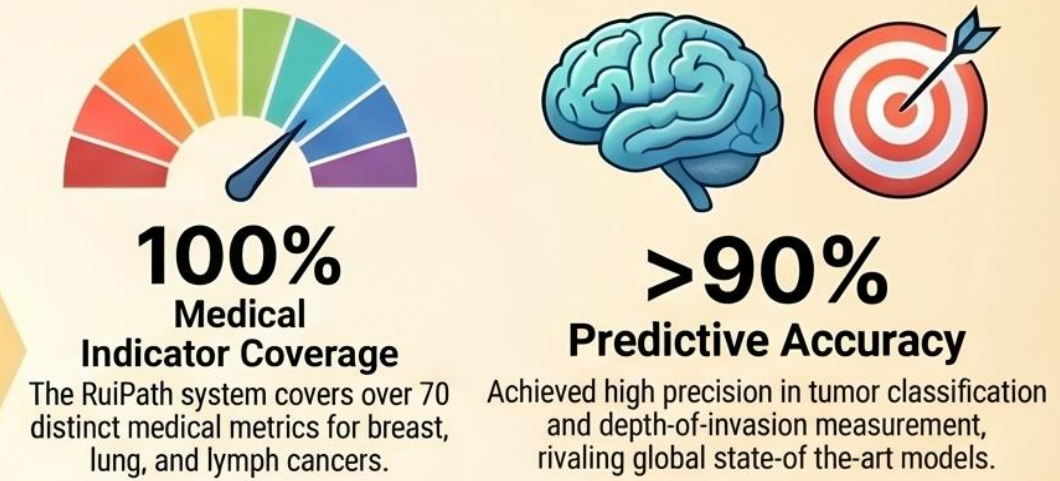
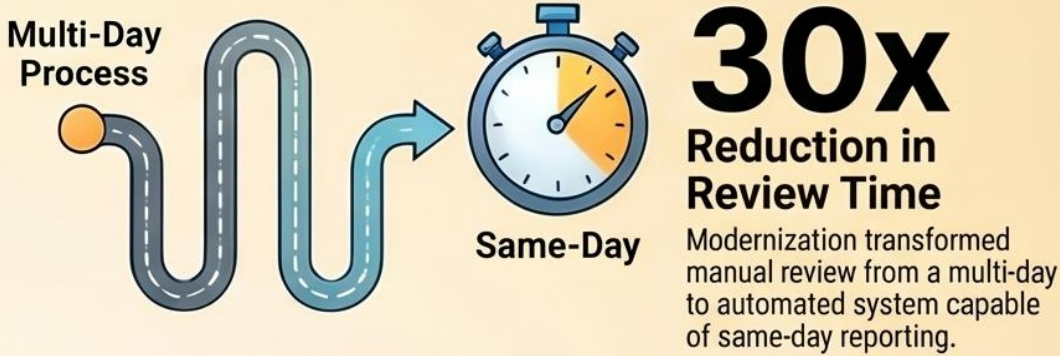
Search



RAG-Based Knowledge Search

Developed a conversational QA system for pathologists using LangChain and GauseDB to query clinical notes and medical textbooks alongside image data.

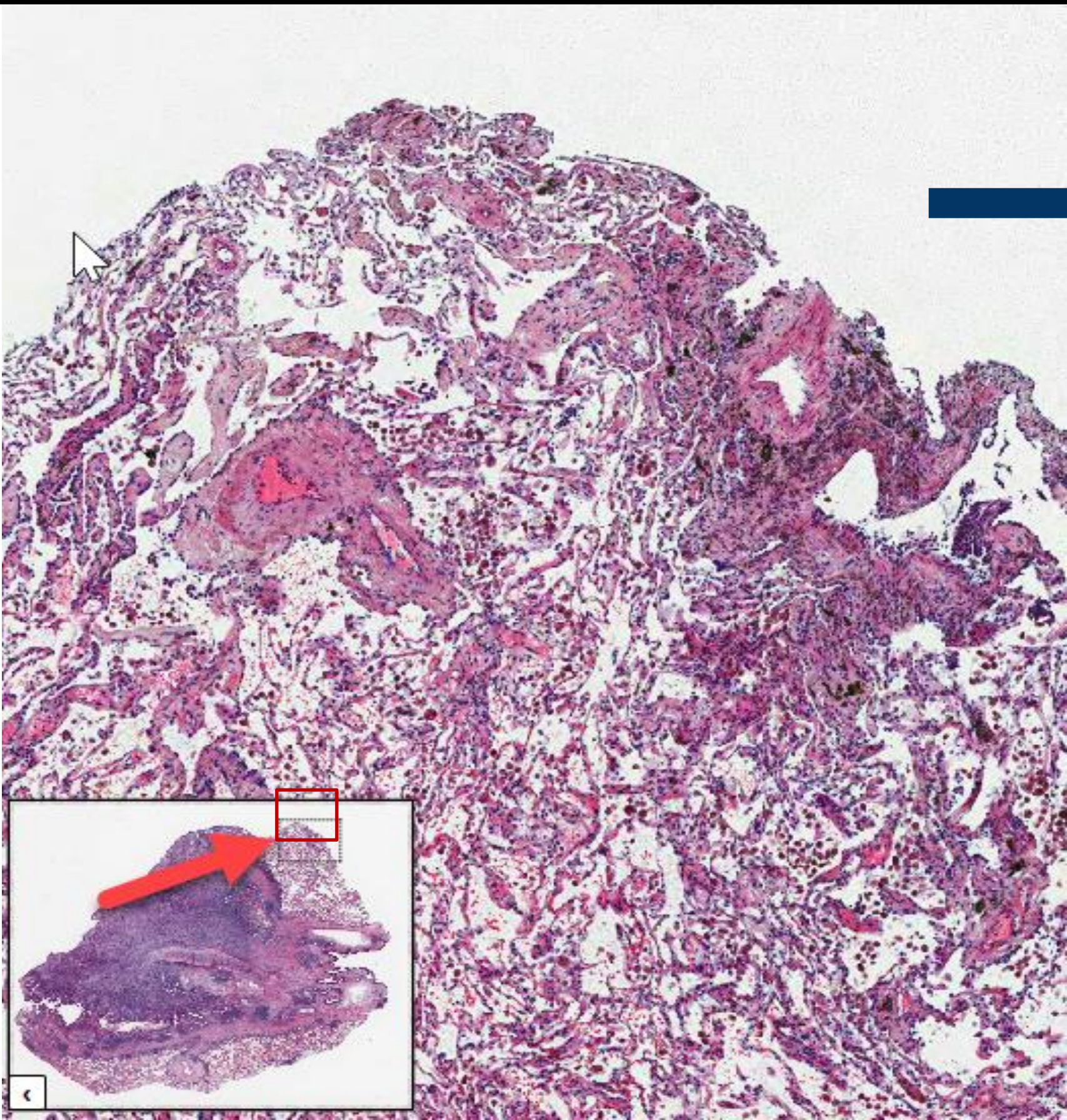
The Value Created: Clinical & Operational Impact



Performance Comparison: RuiPath vs. Global Benchmarks

Task Category	Performance Metric	RuiPath Result	Benchmark Comparison
Clinical Indicators	Coverage Rate	100%	70+ distinct metrics
Tumor Classification	Accuracy	>90%	World-class performance
Research Benchmarks	AUC/ACC/F1	7 Tasks at Global Top	Comparable to Virchow2
Report Turnaround	Cancer Staging	Same-Day	Previously several days

this is a 40x zoom-in image of 1 cm² specimen under digital camera, spotting tumor cells is truly looking for a needle in a haystack, there is AI, but ...

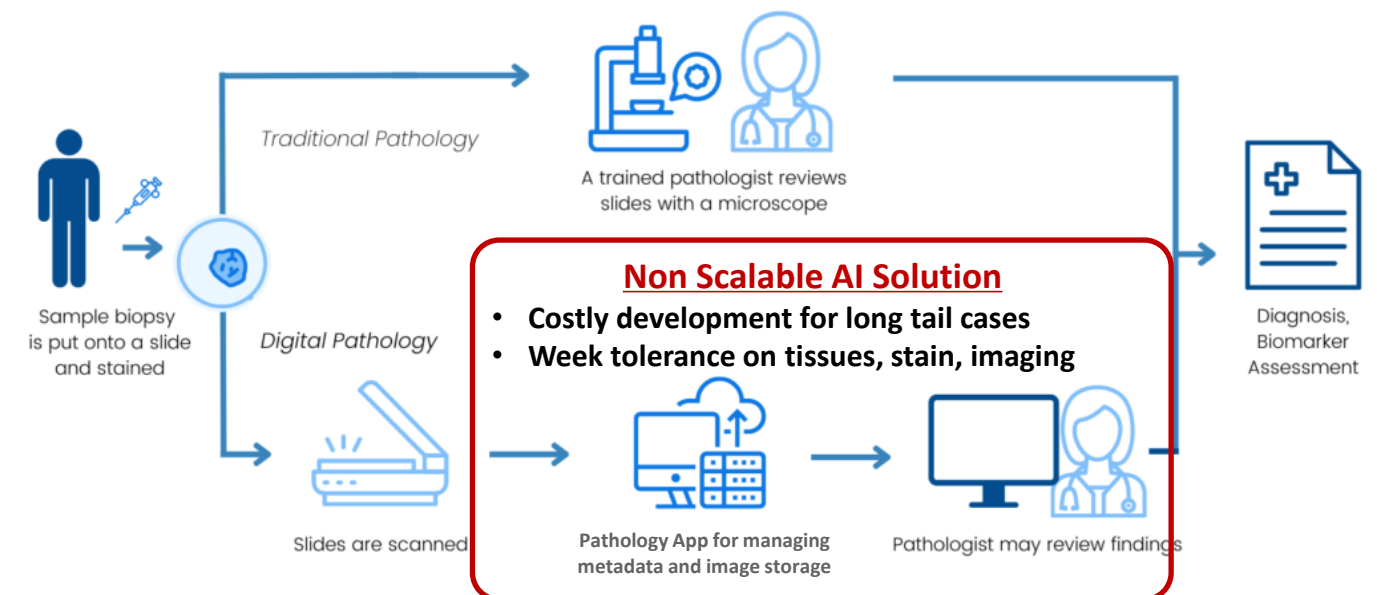


Paint-points

Digital Pathology is NOT Scalable

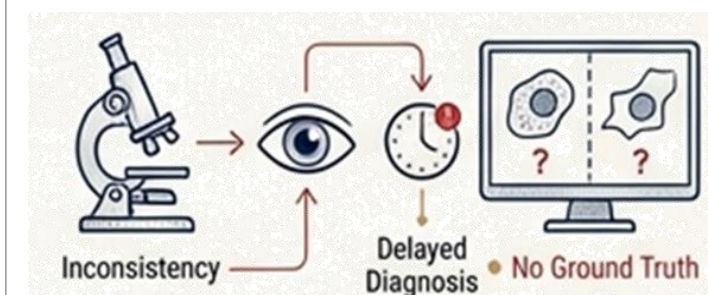
It only create high resolution images, traditional computer vision is **costly** but **weak**

A computer vision model needs thousands of images manually labelled at the cell level for each cancer type and each tissue. Each label costs CNY 2,000 because only a medical specialist can do the work. So, adding just one new feature costs at least CNY 2 million. But unfortunately, there are hundreds features per cancer, so ...



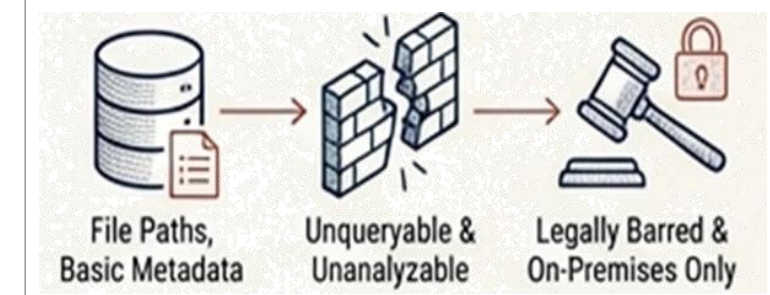
Business Constrains

- 100% reliance on manual review either on big screen or under microscope;
- Tedious, prone to error and inconsistency, cause delay diagnoses.
- Run out of doctors, 10 years to train a certified pathologist (demand > supply)

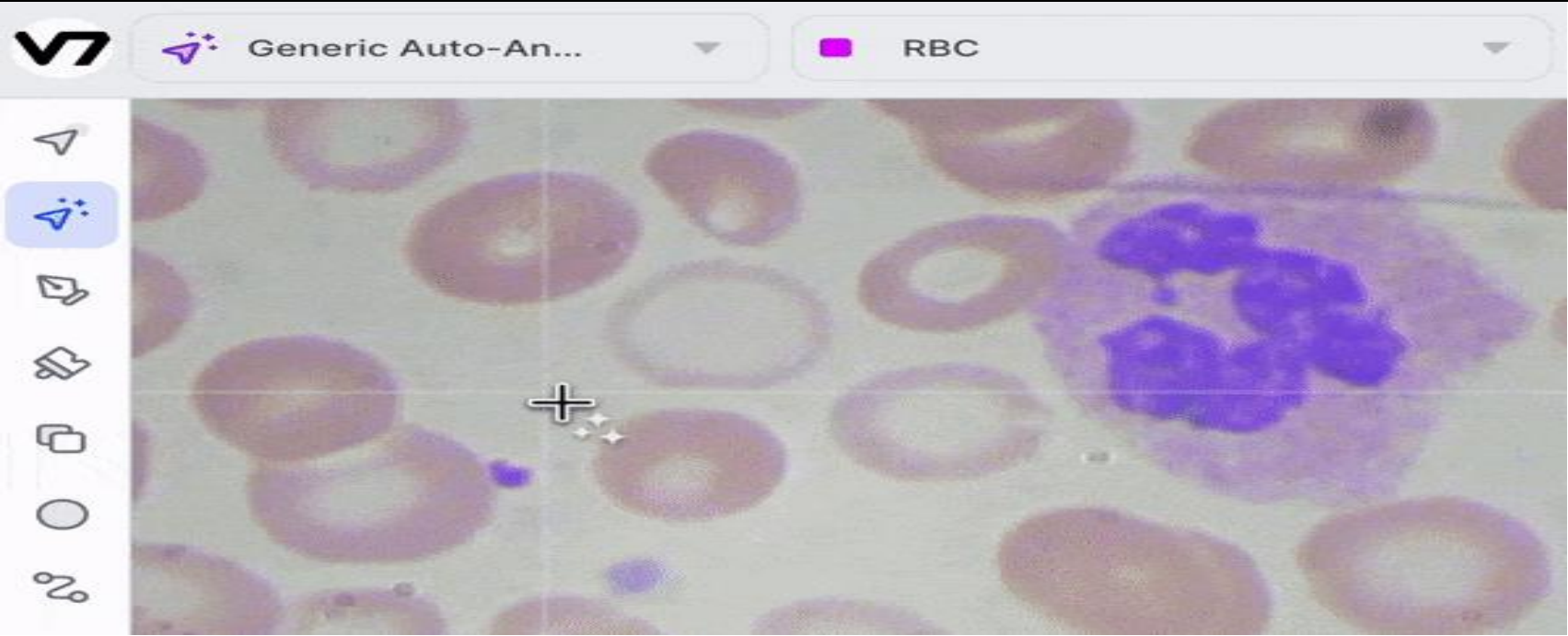


Technology Constrains

- Legacy database stored only metadata and file paths to raw images.
- Extended modules for analytics is week and costly, the hospital just drop those features.
- Data was un-queryable, un-analyzable, and kept for compliance only



Legacy AI (Computer Vision Model) helps doctor performing labeling with intelligent boundary detection, but there are still thousands of those tasks per tissue.



AI Solution

Vision Transformer Scales Digital Pathology

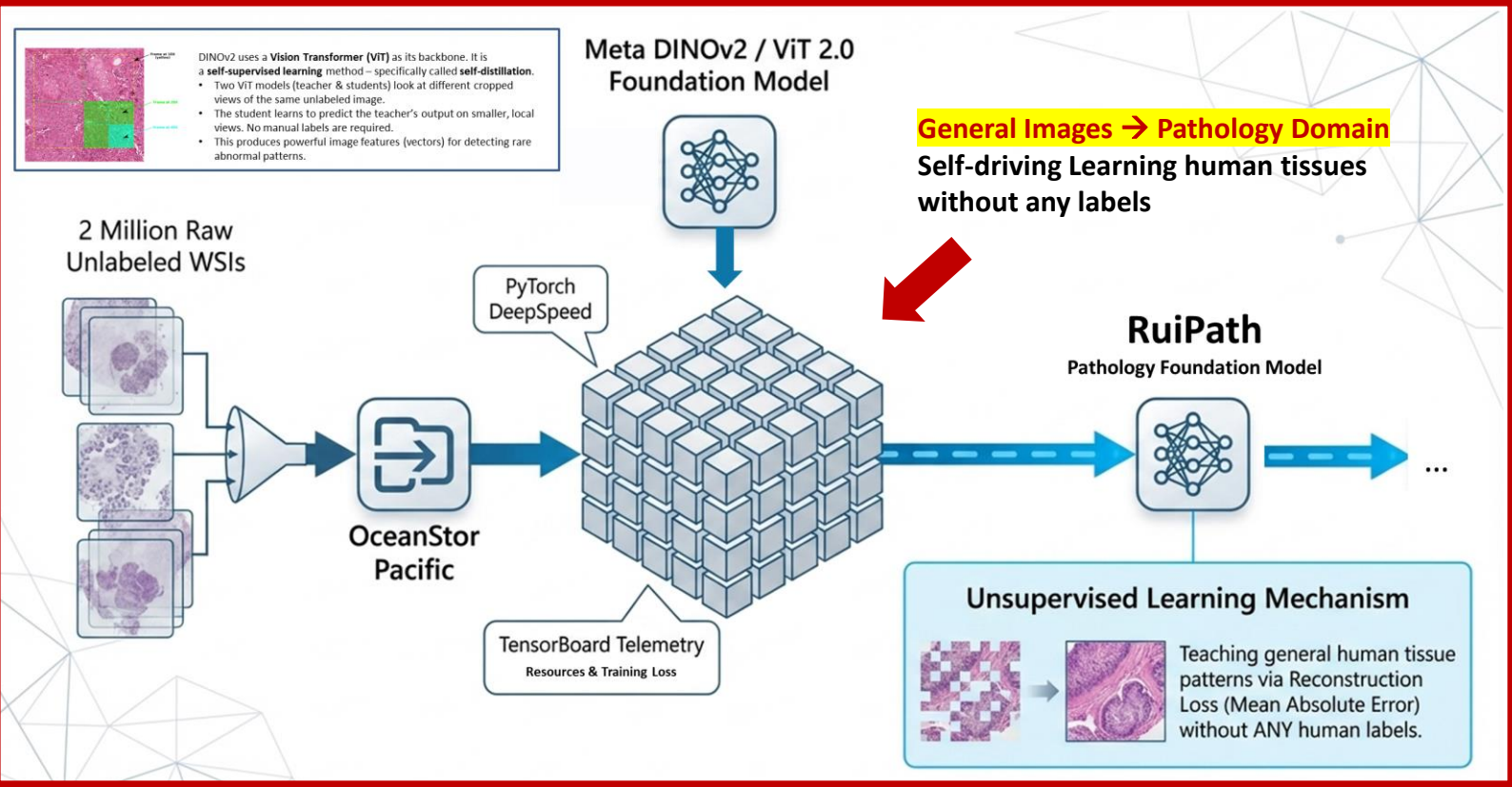
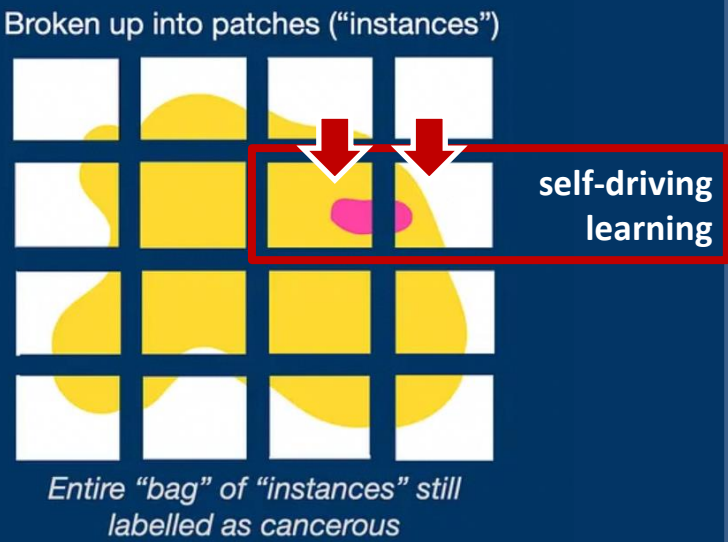
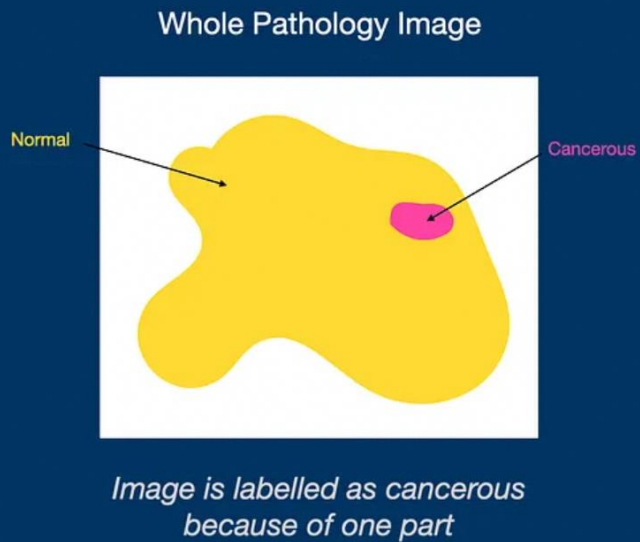
Database Modernization is about building the AI data platform around DBs



From AI assisted labelling to large scale self-learning

Modern Vision Transformer (ViT) empowers Multi-instance learning at Scale

Multi-instance learning (MIL) treats a whole slide image as a “bag” of smaller patches and, using only image-level labels (e.g., “contains tumor” or “normal”), **MIL can learn which patches are most indicative of tumor cells** — without requiring expensive cell-level annotations.



We fine-tuned a vision transformer using millions of whole slide images to build a foundation model for clinical pathology, make **diagnostic intelligence self-driving**

Feature	Legacy AI	Modern GenAI
Database Engine	X Single-instance PostgreSQL X Storing metadata and image path	☺ Distributed GaussDB (Relationship SQL + Vector Embedding) + OceanStor Pacific/S3
Analytics	X tradition AI/ML Dropped due to high cost to develop and week usage	☺ Easy to scale MIL models powered by VisionTransformer foundation model
Data State	X Archive only “dark data” X Unstructured inaccessible	☺ Vectorized Unstructured data for query and search, ready for analytics.
Clinical Workflow	X 100% Manual in digitalized computer screen & microscopic	☺ AI co-pilot with Image Diagnostic Agents & Clinical Knowledge Q&A (RAG)
Turnaround Time	X Multi-day Reporting	☺ 15 Mins end to end reporting Time

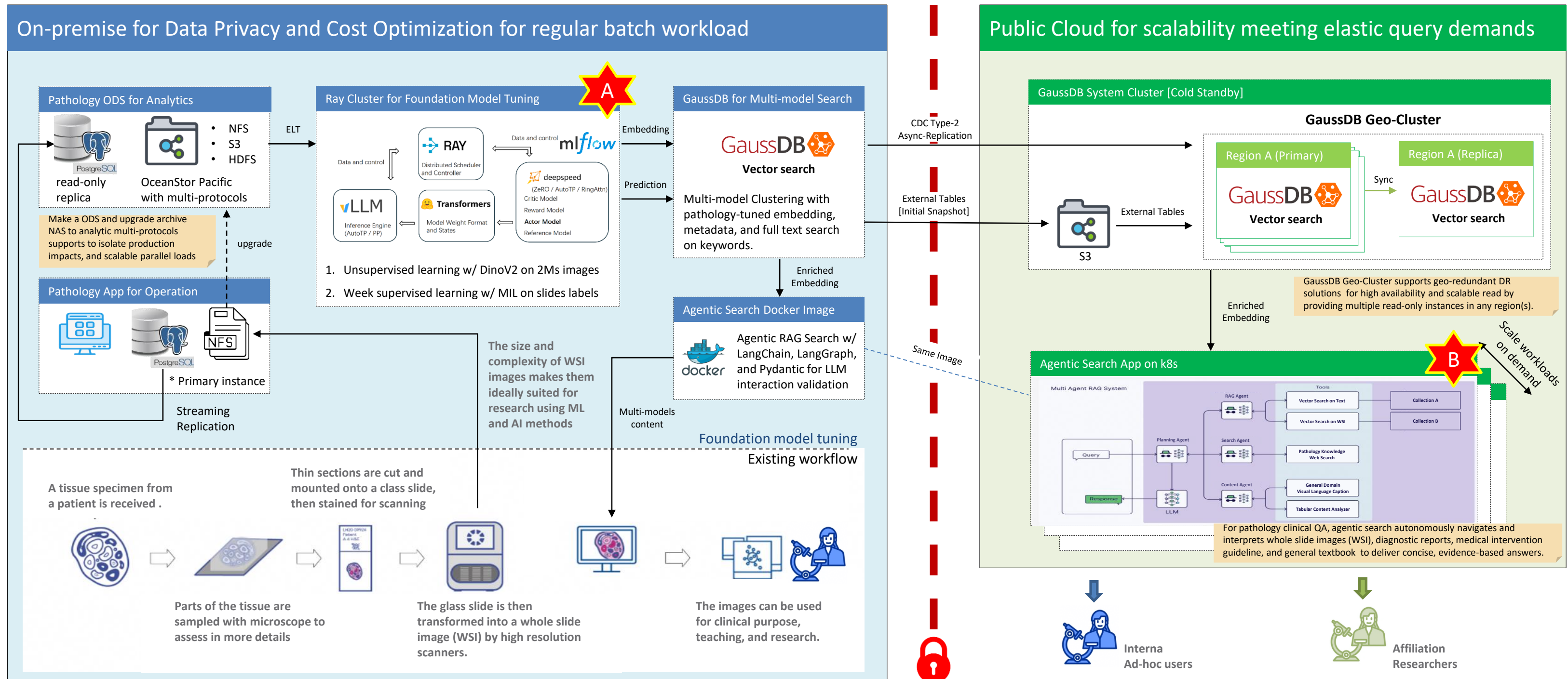
Architecture Overview - Hybrid Cloud Solution

Keeping Pathology Foundation Models and Patient Data on premise, and empowers Queries Scale at Cloud

Stable Workloads With Secure Data Boundary
(Regulatory Compliance – patient related data are not allowed in public cloud)



Scalable Workloads For Meeting Elastic Clinical Demands
(Dynamic Scalability - surge workloads at pandemic or public health crisis)

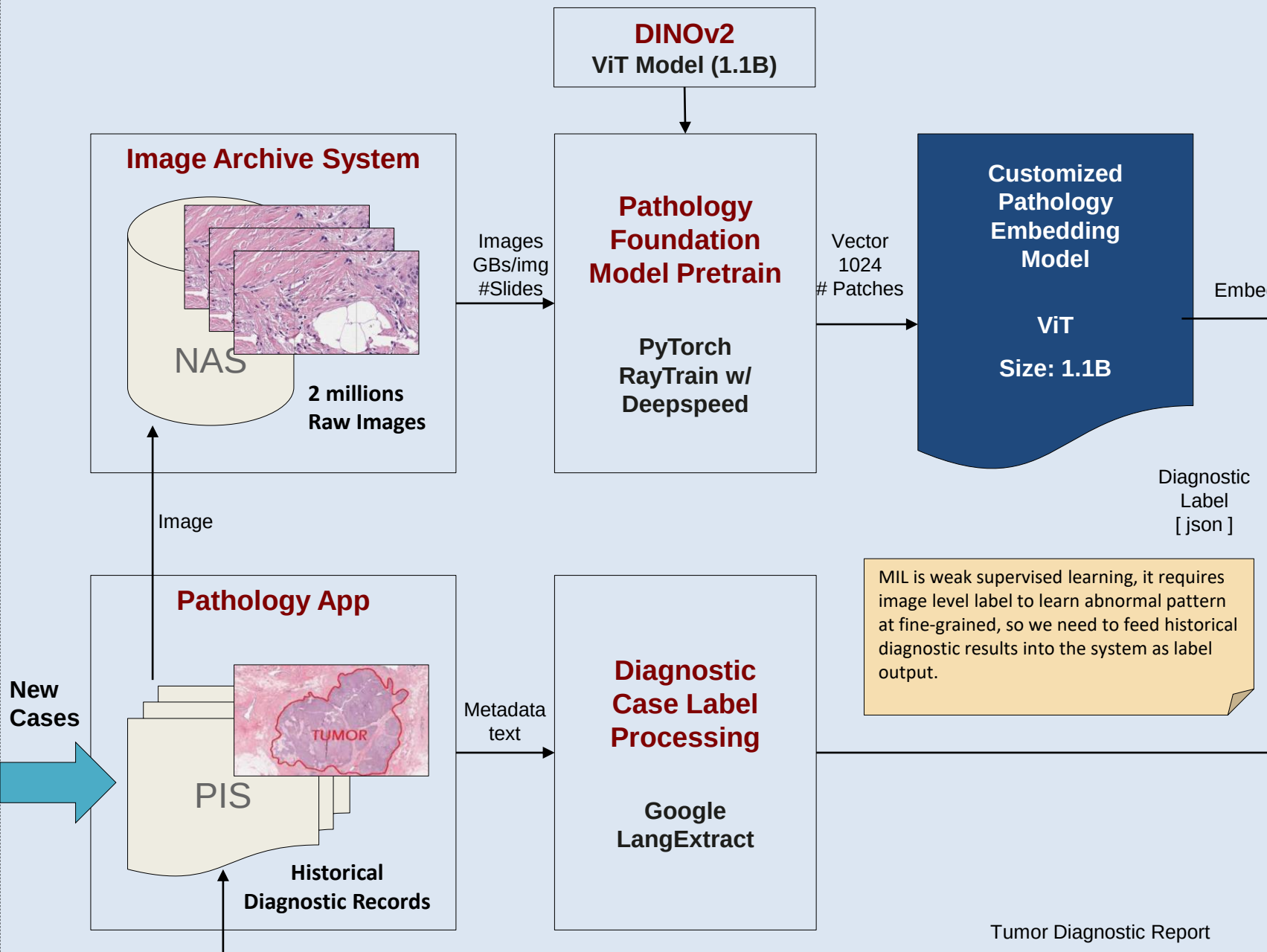


Core Tasks | Terms: { ODS: Operation Data Store; WSI: Whole Slide Image }

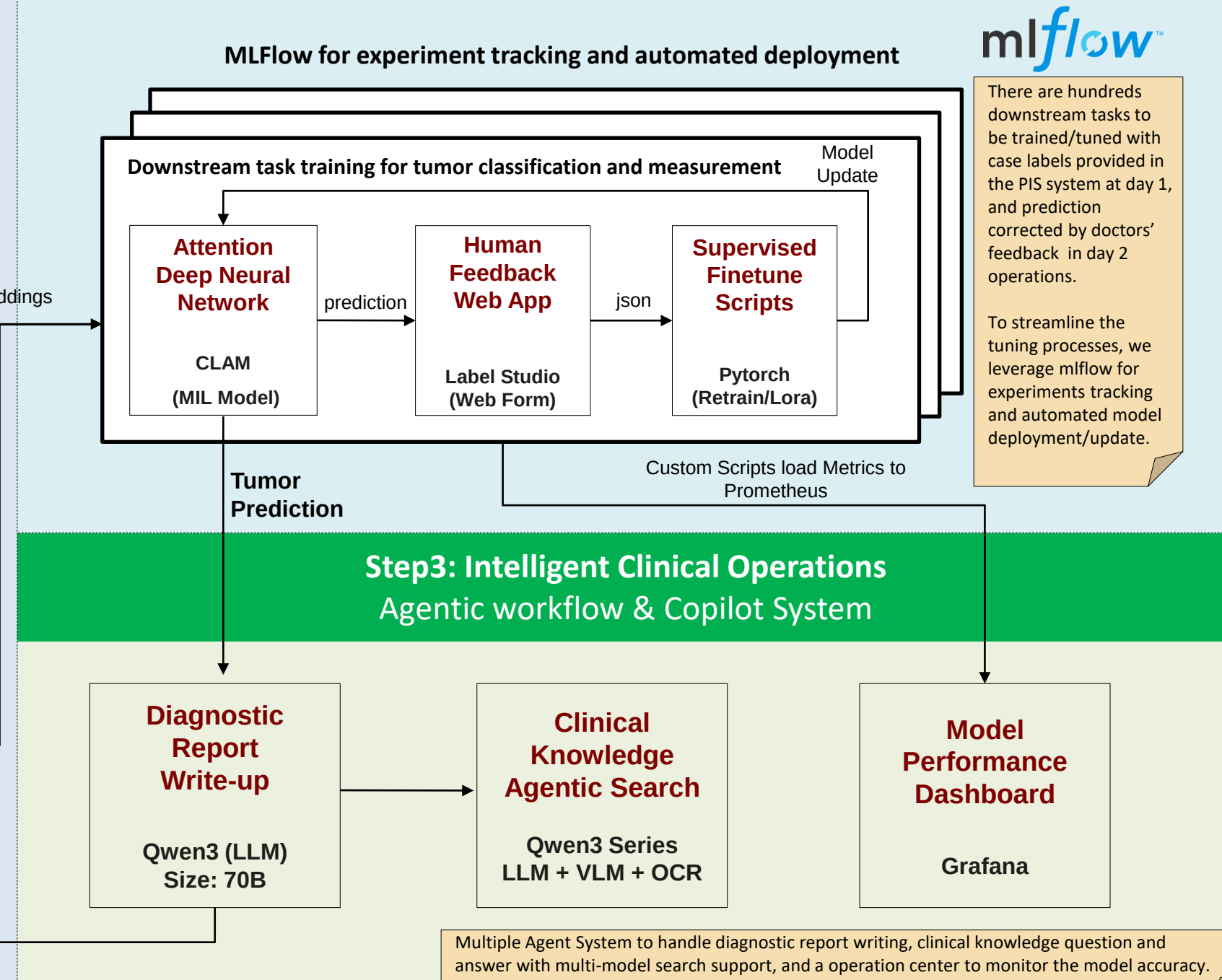
GenAI Core Task A: Model Development for Pathology Foundation Model

Modern Vision Transformer (ViT) empowers Multi-instance learning at Scale, learning image pattern with existing case labels

Step 1: Foundation Model Training (Once) Model Domain Adaptation System

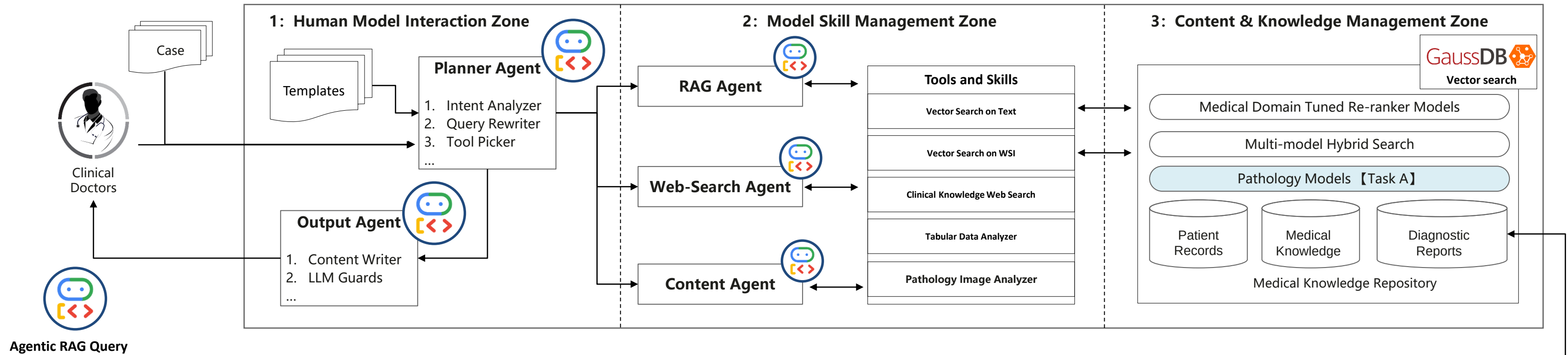


Step 2: Multi-instance Learning (Continuous) Feedback Learning System

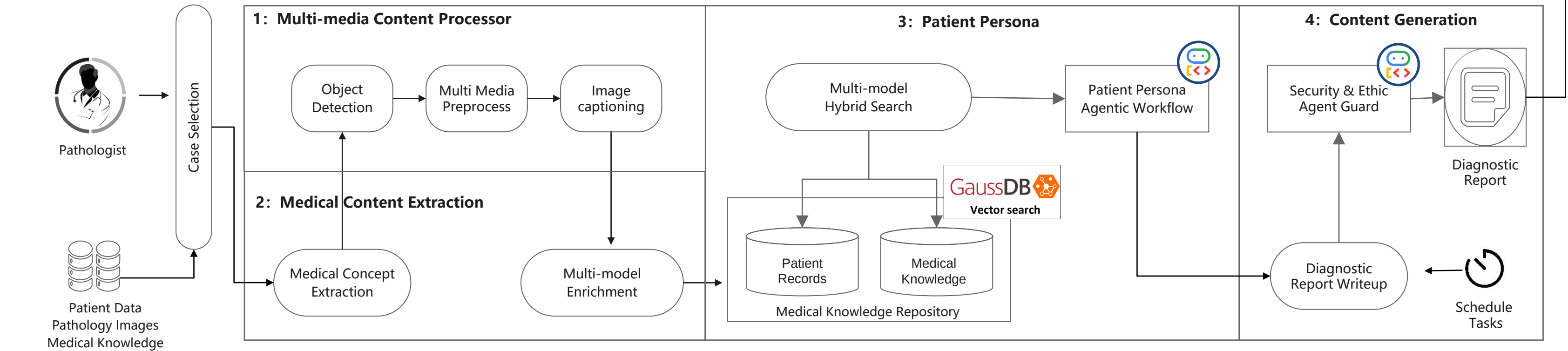


GenAI Core Task B: Multi-models Clinical Knowledge Q&A (Agentic RAG)

Agentic Search on Multi-modal data empowers accuracy contextual rich



Data Ingestion & Schedule Jobs



Outcome

Measurable Business Impacts and Outstanding Research Output

Business Impacts



Production Status:

Deployed for Breast, Lung, and Lymph cancers.



Comprehensive Scope:

Tracking >70 distinct clinical metrics.

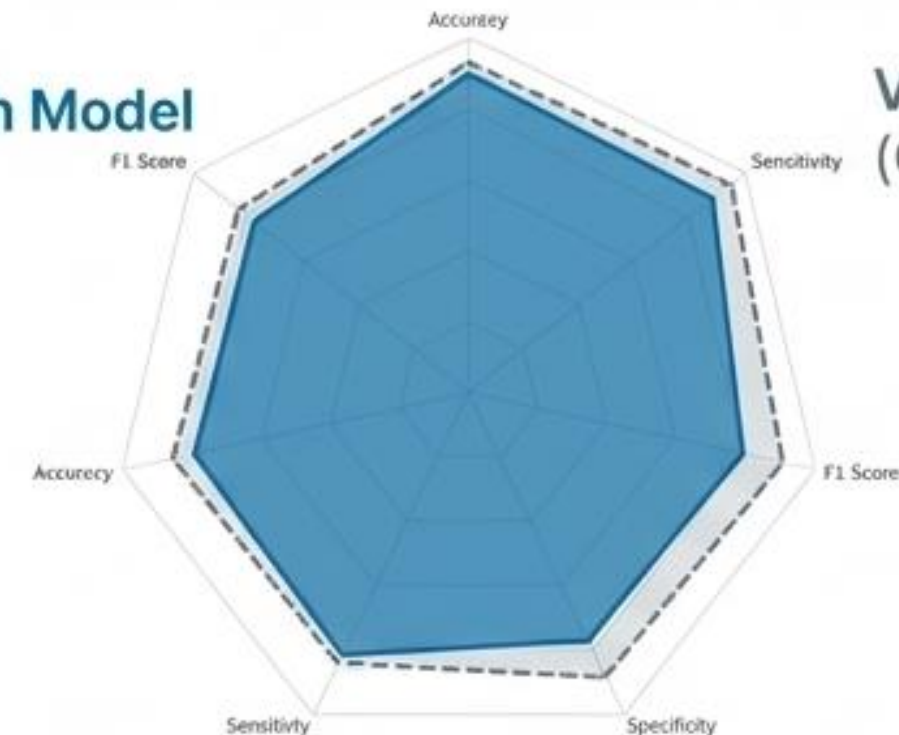


Clinical Precision:

Achieves >90% predictive accuracy.

Research Achievement

RuiPath Model



Virchow2 (Global SOTA)

- Evaluated against **12** mainstream public datasets.
- Tested across **14** auxiliary diagnostic tasks.

Result: 7 tasks achieved world-class performance (AUC, ACC, F1 score) exactly matching the global state-of-the-art Virchow2 model.

My Roles and Responsibilities

Engineering Leadership in High-Stakes Environments

AI Data Platform Solution Architect

-  - Architect and design the data platform infra on Ray Trainer and vllm
-  - Support Foundation Model Training and lead system integration with downstream tasks

Data & Model Engineers (Algorithms & Code)

- Design and implement the multi-instance learning and (MIL) and hierarchical clustering solution.
- Lead the AI&ML solution integration with GaussDB Vector Search

RAG Technical Experts (Research Direction)

- Lead the RAG solution overall architecture
- Build the agentic search solution with all LangChain based orchestration vector applications

Chiwai's Experience - Value Proposition

Delivering compliant, production-grade AI transformations across healthcare, finance, and energy by bridging the gap between legacy infrastructure and generative intelligence.